Name:

Enrollment No:

BML MUNJAL UNIVERSITY
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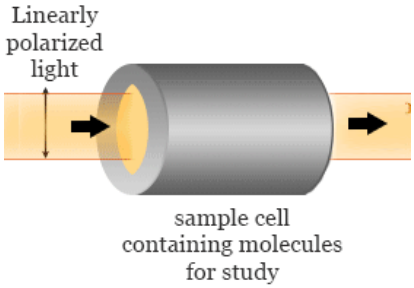
END TERM EXAMINATIONS – DECEMBER 2024
COURSE: PHY1101 – ENGINEERING PHYSICS

PROGRAMME: B.TECH. CSE/ME/ECOM **SEMESTER: I**
DURATION: 2 HRS. **MAX. MARKS:40**

Instructions:

- All Questions are compulsory.
- There is no overall choice.
- Start each answer on a fresh page and number your answers clearly. Answer all parts of the same Question together and in sequence.
- Use of scientific calculator is permitted.

Q.	Section A (Total 15 Marks)	Marks	CO & BL
1.	What is specific rotation? Give its formula (describing each term in formula) along with unit. Explain what happens to electric field vector when linearly polarized light passes through sucrose solution. Draw the figure provided below in your copy. Mark the electric vector oscillations in the radiation emitted out of the sample chamber. 	[0.5 + 0.5 + 0.5 + 0.5]	CO1 &L3
2.	Write differential form of four Maxwell's equation in free space.	[2]	CO1 &L1
3.	Sketch the miller planes in simple cubic crystals represented by miller indices i. (100), ii. $(\bar{3}\bar{2}0)$.	[1+1]	CO1 &L3
4.	The values of refractive indices of e- and o- rays for calcite are 1.48641 and 1.65836 respectively. Determine the thickness of calcite plate (in mm) required to produce phase retardation of 90° when incident light radiation is obtained from a sodium light source.	[3]	CO1 &L3
5.	Define (i) gradient of a scalar function (ii) divergence & curl of a vector function.	[1+2]	CO2 &L1
6.	The band gap of a semiconductor material is 0.7eV. Find out the wavelength (in Å) of the light photon which can just send an electron from valence band to conduction band.	[3]	CO1 &L3

	SECTION B (Total 25 Marks)	Marks	CO & BL
7.	Consider Newton's Ring setup and derive the expression for wavelength along with required labelled diagrams.	[10]	CO1 &L2
8.	Discuss the mechanisms of absorption, spontaneous emission and stimulated emission, providing the formulae of their respective probabilities (describing each	[3+2]	CO1 &L2

parameter) in terms of Einstein's A and B coefficients. Write Einstein's relations in the state of thermal equilibrium.

9.	Derive wave equation for E and B when electromagnetic wave is propagating in free space.	**[5]**	**CO2 &L2**
10.	Derive an expression for the effective mass of an electron and plot a labelled graph to represent its variation.	**[3+2]**	**CO1 &L3**